

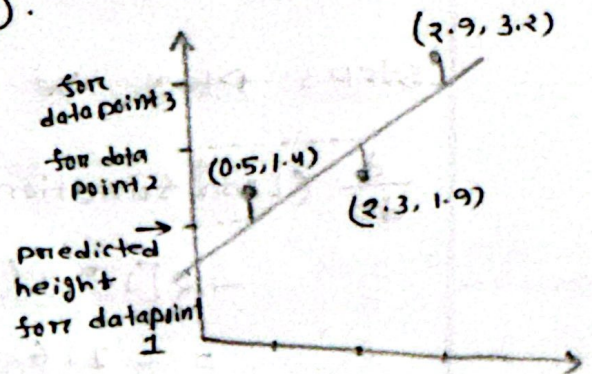
TOPIC NAME: Gradient Descent

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Gradient Descent to find m and c :

Step 1: Taking Derivative of the Loss Function in terms of Parameters (m and c both here).



$$\text{Loss function (SSR)} = [1.4 - (m \times 0.5 + c)]^2 + [1.9 - (m \times 2.3 + c)]^2 + [3.2 - (m \times 2.9 + c)]^2$$

$$\frac{d}{dc} (\text{Loss function}) = -2[1.4 - (m \times 0.5 + c)] - 2[1.9 - (m \times 2.3 + c)] - 2[3.2 - (m \times 2.9 + c)]$$

$$\frac{d}{dm} (\text{Loss function}) = \frac{d}{dm} [1.4 - (m \times 0.5 + c)]^2 + \frac{d}{dm} [1.9 - (m \times 2.3 + c)]^2 + \frac{d}{dm} [3.2 - (m \times 2.9 + c)]^2$$

$$= 2[1.4 - (m \times 0.5 + c)](-0.5) + 2[1.9 - (m \times 2.3 + c)](-2.3) + 2[3.2 - (m \times 2.9 + c)](-2.9)$$

$$= -2 \times 0.5[1.4 - (m \times 0.5 + c)] - 2 \times 2.3[1.9 - (m \times 2.3 + c)] - 2 \times 2.9[3.2 - (m \times 2.9 + c)]$$

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Step 2: Pick a random value for the parameters (here m and c).

Let $m=1$ and $c=0$.

Step 3: Plug the parameter values in derivatives

$$\begin{aligned}\frac{d}{dc} (\text{Loss function}) &= -2[1.4 - (1 \times 0.5 + 0)] \\ &\quad - 2[1.9 - (1 \times 2.3 + 0)] - 2[3.2 - (1 \times 2.9 + 0)] \\ &= -1.6\end{aligned}$$

$$\begin{aligned}\frac{d}{dm} (\text{Loss function}) &= -2 \times 0.5[1.4 - (1 \times 0.5 + 0)] \\ &\quad - 2 \times 2.3[1.9 - (1 \times 2.3 + 0)] - 2 \times 2.9[3.2 - (1 \times 2.9 + 0)] \\ &= -0.8\end{aligned}$$

Step 4: Calculate step size of parameters (m and c)

$$\text{step size } (c) = \frac{d}{dc} \times \text{learning rate}$$

$$= -1.6 \times 0.01 = -0.016$$

$$\text{step size } (m) = \frac{d}{dm} \times \text{learning rate}$$

$$= -0.8 \times 0.01 = -0.008$$

Previous learning rate will not work this time.

Gradient descent is sensitive to learning rates.

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Step 5: Calculate the parameters (c and m)

New intercept, $c_{\text{new}} = c_{\text{old}} - \text{step size}$

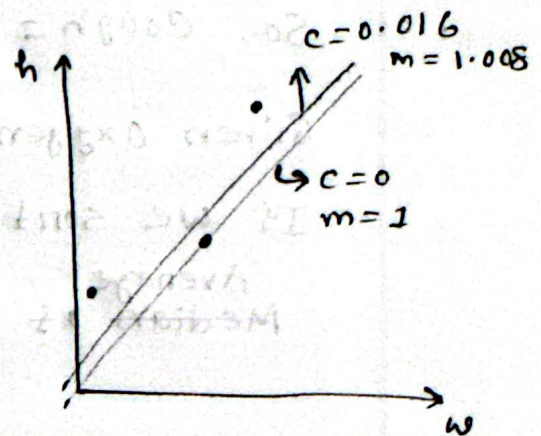
$$= 0 - (-0.016) = 0.016$$

New slope, $m_{\text{new}} = m_{\text{old}} - \text{step size}$

$$= 1 - (-0.008) = 1.008$$

Now we will repeat this process until we reach our termination condition.

Step 1
Step 2
Step 3
Step 4
Step 5



$$c_{\text{new}} = c_{\text{old}} - \frac{d}{dc} (\text{Loss function}) \times \alpha$$

$$m_{\text{new}} = m_{\text{old}} - \frac{d}{dm} (\text{Loss function}) \times \alpha$$